# ATLAS: A Robust Hybrid Pipeline for Temporal Image Change Detection under Camera Motion and Illumination Change

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***Abstract*—Finding what has changed between two images of the same scene taken at different times is central to inspection, monitoring and remote sensing. Naive pixel differencing fails as soon as the camera moves by a few pixels or the illumination shifts, because such nuisance changes alter far more pixels than a real change does. We present ATLAS, a training-free pipeline that (i) registers the later image to the earlier one with ORB features and a RANSAC homography, (ii) corrects illumination with a robust median/MAD transfer in LAB colour space, (iii) fuses a pixel-difference map with a structural-dissimilarity (1−SSIM) map, and (iv) thresholds the fused map with Otsu's method under a tunable lower bound, followed by morphological clean-up. We compare ATLAS with pixel differencing, SSIM and a fixed-weight hybrid, and ablate each component, on a seven-scenario synthetic benchmark with exact ground truth and separate validation and test splits. On 210 held-out pairs ATLAS reaches a mean F1 of 0.933, against 0.797 for pixel differencing, 0.702 for the fixed hybrid and 0.620 for SSIM (paired Wilcoxon tests, $p < 10^{-22}$), and the ranking replicates on a second, independently generated test set. Registration yields the largest single gain, and the components are complementary. Contrary to our initial hypothesis, fusing pixel and SSIM maps with a fixed weight does not outperform pixel differencing. Tests on three real phone photographs, which lack ground truth, show where the method breaks: strong parallax and large exposure differences violate its global geometric and photometric assumptions. The pipeline is implemented in Python with a FastAPI service and an interactive web interface.**

***Index Terms*—change detection, image differencing, structural similarity, image registration, illumination normalisation, Otsu thresholding, ablation study**

## I. Introduction

Comparing two images of the same scene taken at different times in order to find what has changed is a basic operation in infrastructure inspection, security monitoring, environmental assessment and remote sensing [1], [2]. The task sounds simple, but a pixel-wise subtraction of two photographs rarely isolates the changes of interest. Real image pairs differ for many reasons unrelated to the scene content: the camera moves by a few pixels, exposure and white balance vary, sensors add noise, and shadows shift. These nuisance changes affect far more pixels than a small object appearing or disappearing, so a detector that cannot separate them from semantic changes produces masks dominated by false alarms.

Deep Siamese networks [3] and attention-based models trained on annotated datasets such as LEVIR-CD [4] define the state of the art on benchmark data, but they require labelled image pairs, GPU training and careful domain matching. For a small team, a classroom or an embedded setting, a classical training-free pipeline is attractive because it is cheap, interpretable and easy to debug. What is less clear is *which parts of such a pipeline matter*. This paper asks three questions:

- **RQ1.** Does fusing pixel differences with structural dissimilarity (SSIM) improve over either map alone?
- **RQ2.** How much do registration, photometric normalisation and adaptive thresholding each contribute, alone and combined?
- **RQ3.** Where does the resulting pipeline fail on real photographs?

Our contributions are:

1) **ATLAS**, a four-stage, training-free pipeline in which every stage can be switched on or off, so that each stage's contribution can be measured.
2) A reproducible **synthetic benchmark** of seven scenarios with exact ground-truth masks, separate validation and test splits, and a documented parameter-selection protocol.
3) A **component ablation with paired significance tests** and a replication on a second, independently generated test set.
4) An account of **negative results and failure cases**: fixed fusion of pixel and SSIM maps does not beat pixel differencing, and real photographs with parallax and large exposure differences expose the limits of global models.

The code is written in Python with OpenCV and scikit-image and is exposed through a FastAPI service and a Next.js interface used for the real-photo demonstrations in Section VI. Code, seeds and result files: https://github.com/sadSanta-07/ATLAS.

## II. Related Work

*Classical change detection.* Image differencing, ratioing, regression and thresholding are the oldest tools for detecting change between co-registered images and are surveyed in [1], [2]. These surveys identify imperfect registration and radiometric differences between acquisitions as central difficulties, which is exactly what our nuisance scenarios emulate.

*Structural similarity.* SSIM [5] compares local luminance, contrast and structure and was designed for image quality assessment. Its complement $1 - \mathrm{SSIM}$ is a natural candidate for a structural change score. As our experiments show, being a local, structure-based measure it reacts strongly to small spatial offsets.

*Registration and radiometric correction.* Feature-based registration with ORB [6] and RANSAC [7] estimates a homography, which models the image motion exactly only for planar scenes or pure camera rotation [8]. Statistical colour transfer [9] matches the first- and second-order statistics of one image to another; we use a robust median/MAD variant so that a changed region biases the correction less.

*Thresholding.* Otsu's method [10] selects the threshold that maximises between-class variance. It presupposes two populations of pixels, an assumption that fails when nothing has changed, which motivates the lower bound we introduce in Section III.

*Learning-based methods.* Fully convolutional Siamese networks [3] and spatial-temporal attention models with the LEVIR-CD dataset [4] learn the notion of "relevant change" from annotations. We do not compare against learned models, and our conclusions concern classical pipelines only.

## III. The ATLAS Pipeline

### A. Problem statement

Given two colour images $I_1$ (earlier) and $I_2$ (later) of the same scene, the goal is a binary mask $M \in \{0,1\}^{H \times W}$ in the coordinate frame of $I_1$ with $M(x,y) = 1$ where the scene content has changed. *Semantic* changes (objects appearing or disappearing) should be detected; *nuisance* changes (camera motion, illumination, sensor noise) should not. Fig. 1 summarises the pipeline.

### B. Preprocessing and registration

$I_1$ is downscaled (never upscaled) so that its longest side is at most 1024 px, and $I_2$ is resized to the size of $I_1$. We detect up to 3000 ORB features in both images [6], match them with a brute-force Hamming matcher with cross-checking, keep the best 60% of the matches, and estimate a homography $\mathbf{H}$ with RANSAC (reprojection threshold 3 px) [7]. The estimate is accepted only if at least 25 inliers support it; otherwise $I_2$ is left unchanged, so the stage degrades gracefully. The warped image is resampled bilinearly with replicated borders.

### C. Robust photometric normalisation

To compensate global exposure and colour differences, $I_2$ is converted to CIE LAB and, per channel $c$, its distribution is mapped onto that of $I_1$:

$$\tilde{I}_2^c = \left(I_2^c - m_2^c\right) \frac{s_1^c}{s_2^c} + m_1^c, \tag{1}$$

where $m^c$ is the median and $s^c = 1.4826\,\mathrm{MAD}$ the scaled median absolute deviation of channel $c$. Median/MAD replace the mean and standard deviation of classical colour transfer [9] so that a moderately sized changed region does not drag the statistics. The model is global and assumes that most pixels are unchanged.

### D. Score maps and fusion

Both images are smoothed with a Gaussian ($\sigma = 1$). The pixel score is the mean absolute RGB difference,

$$D_\mathrm{p}(x,y) = \frac{1}{3} \sum_{c \in \{R,G,B\}} \frac{\left| I_1^c(x,y) - \tilde{I}_2^c(x,y) \right|}{255}, \tag{2}$$

and the structural score is $D_\mathrm{s} = \mathrm{clip}(1 - \mathrm{SSIM}, 0, 1)$, with SSIM computed on grayscale images with a Gaussian window ($\sigma = 1.5$) [5]:

$$\mathrm{SSIM} = \frac{(2\mu_1\mu_2 + C_1)(2\sigma_{12} + C_2)}{(\mu_1^2 + \mu_2^2 + C_1)(\sigma_1^2 + \sigma_2^2 + C_2)}. \tag{3}$$

The fused score is

$$D_\mathrm{h} = \alpha\, D_\mathrm{p} + (1 - \alpha)\, D_\mathrm{s}, \qquad \alpha \in [0,1]. \tag{4}$$

### E. Adaptive decision with a floor

With a fixed threshold $\tau$, $M = \mathbf{1}[D_\mathrm{h} > \tau]$. The adaptive variant uses Otsu's threshold [10], but with a lower bound:

$$\tau = \max\left(\tau_\mathrm{Otsu}(D_\mathrm{h}),\, \tau_\mathrm{min}\right), \qquad M(x,y) = \mathbf{1}\left[D_\mathrm{h}(x,y) > \tau\right]. \tag{5}$$

Otsu's method always returns a split, even when the score map contains only noise; without a floor it then labels part of the noise as "change". We observed exactly this on unchanged image pairs (Section V-D), and $\tau_\mathrm{min}$ is selected on the validation split like every other parameter.

### F. Post-processing

The mask is cleaned by a morphological opening followed by a closing with an elliptical $5 \times 5$ kernel, and connected components (8-connectivity) smaller than 50 px are discarded. The remaining components are reported as regions with bounding box and area.

## IV. Experimental Setup

### A. Synthetic benchmark

Real image pairs with exact change masks and controlled nuisance factors are scarce, so we generate them. Each scene is a $512 \times 384$ image built from a smooth random colour field, 70 random circles and rectangles as clutter, and blurred Gaussian texture (which also gives the feature detector something to match). One to three textured objects (ellipses with semi-axes

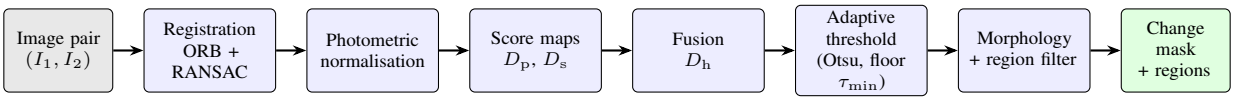



Fig. 1. The ATLAS pipeline. Registration, photometric normalisation and Otsu thresholding can each be disabled, which defines the ablation configurations of Table II.

TABLE I
BENCHMARK SCENARIOS. ALL SCENARIOS ALSO INCLUDE SENSOR NOISE ($\sigma = 2$) IN BOTH IMAGES UNLESS STATED.

| Scenario | Real change | Nuisance in $I_2$ |
|---|---|---|
| object | 1–3 objects appear | none |
| removal | 1–3 objects disappear | none |
| noise | objects appear | noise $\sigma = 10$ |
| brightness | objects appear | gain $1\pm(0.15–0.30)$, bias $\pm25$, horizontal gradient |
| translation | objects appear | shift 2–4 px per axis, rotation $\pm1^\circ$ |
| combined | objects appear | illumination + camera motion + noise $\sigma = 5$ |
| unchanged | none | independent noise only |

of 20–50 px or rectangles of 30–80 px) are inserted with a minimum colour contrast to the background where possible. Gaussian sensor noise ($\sigma = 2$) is added to both images. The ground-truth mask is always expressed in the frame of $I_1$, so that a method facing camera motion must undo it to be correct. Table I lists the seven scenarios.

### B. Splits, tuning protocol and statistics

The *validation* split has 15 pairs per scenario (105 pairs, seed 1000); the *test* split has 30 pairs per scenario (210 pairs, seed 3000). An earlier run on a different test split (seed 2000) is used for replication (Section V-D). Every configuration is tuned on the validation split only, by grid search over $\alpha \in \{0.25, 0.5, 0.75\}$, $\tau \in \{0.05, 0.10, \ldots, 0.60\}$ and, for adaptive configurations, $\tau_\mathrm{min} \in \{0.05, 0.08, 0.10, 0.12, 0.15, 0.20\}$, maximising mean F1 pooled over all validation pairs. The selected parameters are then frozen and evaluated once on the test split. Differences between configurations are tested on the 210 paired per-image F1 scores with the Wilcoxon signed-rank test [11].

### C. Configurations and metrics

Table II defines the four main methods (M1–M4) and three single-component ablations (A1–A3) together with the tuned parameters. We compute pixel-level precision, recall, F1 and IoU per image pair ($\mathrm{IoU} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP} + \mathrm{FN})$) and report the mean and standard deviation over pairs. If both prediction and ground truth are empty (scenario `unchanged`), all four scores are defined as 1; this convention raises the overall average for every method, so we also report averages without that scenario. Runtime is the wall-clock time of the full pipeline after image decoding, on a consumer laptop CPU (single process, no GPU).

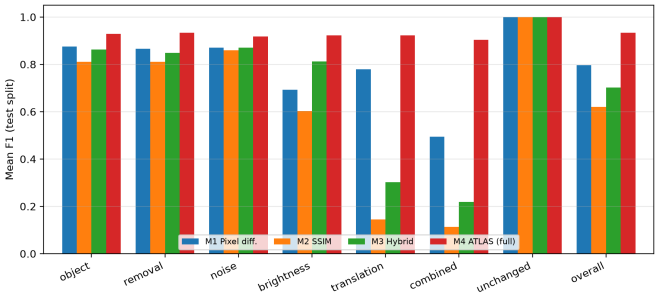



Fig. 2. Mean F1 per scenario for the four main methods (test split, seed 3000).

### D. Implementation

The pipeline uses Python 3.12, OpenCV 5.0, scikit-image 0.26, NumPy 2.5 and SciPy. The detector is a library with a single entry point shared by the experiment scripts and a FastAPI service; a Next.js front end lets a user upload two images, switch between methods, inspect heatmaps and overlays on a comparison slider, and export results. Every run is described by one parameter object, which makes the experiments reproducible from the stored seeds and parameters.

## V. RESULTS

### A. Overall comparison

Table III gives the test-split results. ATLAS (M4) reaches a mean F1 of $0.933 \pm 0.043$ and an IoU of 0.877, ahead of pixel differencing (F1 0.797), the fixed hybrid (0.702) and SSIM (0.620). Paired Wilcoxon tests give mean F1 differences of $+0.136$ against M1, $+0.313$ against M2, $+0.231$ against M3 and $+0.080$ against the strongest ablation A3, with $p < 10^{-22}$ in all four comparisons ($n = 210$). Part of this gain is shared by all methods: because of the `unchanged` convention, the averages are lifted by one scenario in which every method scores 1. Without that scenario, the mean F1 values computed from Table IV are approximately 0.92 (M4), 0.83 (A3), 0.76 (M1), 0.65 (M3) and 0.56 (M2); the ranking does not change.

SSIM alone has high recall (0.877) but the lowest precision (0.584): it marks many pixels that are not part of any real change. Pixel differencing is more precise (0.860) but misses real changes (recall 0.805). ATLAS improves both, with precision 0.912 and recall 0.958, and its F1 spread over pairs is about five times smaller than that of M1 (0.043 versus 0.227), i.e. it is more consistent across conditions.

### B. Per-scenario behaviour

Table IV and Fig. 2 break the results down by scenario.

TABLE II

EVALUATED CONFIGURATIONS AND PARAMETERS SELECTED ON THE VALIDATION SPLIT ($\alpha$: PIXEL WEIGHT; $\tau$: FIXED THRESHOLD; $\tau_{\min}$: OTSU FLOOR; DASHES: NOT APPLICABLE).

| ID | Score map | Registration | Photometric | Threshold | $\alpha$ | $\tau$ | $\tau_{\min}$ | Val. F1 |
|---|---|---|---|---|---|---|---|---|
| M1 (pixel difference) | $D_{\mathrm{p}}$ | – | – | fixed | – | 0.20 | – | 0.799 |
| M2 (SSIM) | $D_{\mathrm{s}}$ | – | – | fixed | – | 0.10 | – | 0.630 |
| M3 (fixed hybrid) | $D_{\mathrm{h}}$ | – | – | fixed | 0.75 | 0.20 | – | 0.708 |
| A1 (hybrid + Otsu) | $D_{\mathrm{h}}$ | – | – | Otsu | 0.75 | – | 0.15 | 0.729 |
| A2 (hybrid + photometric) | $D_{\mathrm{h}}$ | – | ✓ | fixed | 0.75 | 0.15 | – | 0.762 |
| A3 (hybrid + registration) | $D_{\mathrm{h}}$ | ✓ | – | fixed | 0.75 | 0.20 | – | 0.842 |
| **M4 (ATLAS, full)** | $D_{\mathrm{h}}$ | ✓ | ✓ | Otsu | 0.75 | – | 0.12 | 0.936 |

TABLE III

TEST-SPLIT RESULTS (210 PAIRS, SEED 3000), MEAN $\pm$ STANDARD DEVIATION OVER IMAGE PAIRS. RUNTIME IS THE MEAN PER PAIR AT $512 \times 384$.

| Configuration | Precision | Recall | F1 | IoU | Time (ms) |
|---|---|---|---|---|---|
| M1 pixel difference | 0.860 $\pm$ 0.270 | 0.805 $\pm$ 0.151 | 0.797 $\pm$ 0.227 | 0.709 $\pm$ 0.256 | 6 |
| M2 SSIM | 0.584 $\pm$ 0.368 | 0.877 $\pm$ 0.142 | 0.620 $\pm$ 0.352 | 0.537 $\pm$ 0.349 | 54 |
| M3 fixed hybrid | 0.700 $\pm$ 0.359 | 0.846 $\pm$ 0.148 | 0.702 $\pm$ 0.307 | 0.618 $\pm$ 0.329 | 46 |
| A1 hybrid + Otsu | 0.683 $\pm$ 0.364 | 0.917 $\pm$ 0.128 | 0.721 $\pm$ 0.328 | 0.653 $\pm$ 0.352 | 49 |
| A2 hybrid + photometric | 0.706 $\pm$ 0.351 | 0.939 $\pm$ 0.067 | 0.744 $\pm$ 0.309 | 0.674 $\pm$ 0.336 | 70 |
| A3 hybrid + registration | 0.889 $\pm$ 0.156 | 0.851 $\pm$ 0.149 | 0.853 $\pm$ 0.148 | 0.767 $\pm$ 0.185 | 69 |
| **M4 ATLAS (full)** | **0.912 $\pm$ 0.056** | **0.958 $\pm$ 0.056** | **0.933 $\pm$ 0.043** | **0.877 $\pm$ 0.075** | 119 |

TABLE IV

MEAN F1 PER SCENARIO ON THE TEST SPLIT (SEED 3000). BEST VALUE PER COLUMN IN BOLD.

| Configuration | object | removal | noise | brightness | translation | combined | unchanged | overall |
|---|---|---|---|---|---|---|---|---|
| M1 pixel difference | 0.875 | 0.866 | 0.870 | 0.692 | 0.779 | 0.495 | **1.000** | 0.797 |
| M2 SSIM | 0.810 | 0.811 | 0.859 | 0.603 | 0.145 | 0.113 | **1.000** | 0.620 |
| M3 fixed hybrid | 0.862 | 0.849 | 0.871 | 0.812 | 0.302 | 0.219 | **1.000** | 0.702 |
| A1 hybrid + Otsu | 0.930 | 0.931 | **0.922** | 0.753 | 0.274 | 0.236 | **1.000** | 0.721 |
| A2 hybrid + photometric | 0.920 | 0.928 | 0.909 | 0.916 | 0.269 | 0.265 | **1.000** | 0.744 |
| A3 hybrid + registration | 0.862 | 0.849 | 0.871 | 0.812 | 0.868 | 0.707 | **1.000** | 0.853 |
| **M4 ATLAS (full)** | 0.929 | **0.934** | 0.918 | **0.923** | **0.922** | **0.904** | **1.000** | **0.933** |

*Without nuisance* (object, removal) all methods score at least 0.81, and M4 reaches 0.93. *Camera motion* is the most damaging factor for the structural score: SSIM collapses to F1 0.145 in the translation scenario and the fixed hybrid to 0.302, while pixel differencing, helped by the Gaussian pre-smoothing, keeps 0.779. Registration restores the hybrid to 0.868 and M4 to 0.922. *Illumination change* lowers pixel differencing to 0.692; photometric normalisation lifts the hybrid from 0.812 to 0.916. In the *combined* scenario, which stacks both nuisances plus noise, M1, M3 and M2 score 0.495, 0.219 and 0.113, registration alone reaches 0.707, and only the full pipeline exceeds 0.90 (0.904). M4 is the only configuration with F1 $\geq 0.90$ in every scenario.

### C. Ablation

Fig. 3 shows overall F1 with 95% confidence intervals. Starting from the fixed hybrid (0.702), the single components add $+0.019$ (adaptive threshold), $+0.042$ (photometric normalisation) and $+0.151$ (registration), whereas all three together add $+0.231$. The sum of the individual gains ($+0.212$) is slightly smaller than the joint gain, so the components are complementary rather than redundant. This is most visible in the combined scenario, where photometric normalisation alone barely helps (0.265) while photometric normalisation and the adaptive threshold together contribute $+0.197$ on top of registration ($0.707 \rightarrow 0.904$): once the motion is removed, the residual error is dominated by the illumination change.

The adaptive threshold has a second effect that the overall score hides. In scenarios without motion or illumination change it already helps considerably (object $0.862 \rightarrow 0.930$, removal $0.849 \rightarrow 0.931$, noise $0.871 \rightarrow 0.922$), presumably because a per-image threshold copes with the varying contrast of the inserted objects, which a single global $\tau$ cannot.

### D. Replication and the effect of the Otsu floor

We first ran the full experiment on a different test split (seed 2000), with the Otsu floor fixed at 0.05. Table V shows that the ranking of the four methods is identical on both splits. On the `unchanged` scenario, however, M4 only reached F1 0.967 in that first run, i.e. it produced false alarms: with a floor of 0.05, Otsu's method split noise into two groups. We therefore made the floor a tunable parameter, which the validation split set

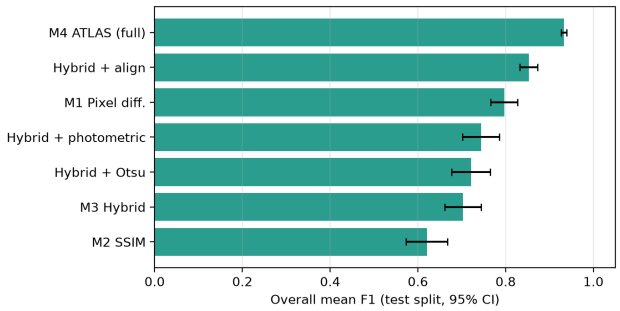



Fig. 3. Overall F1 of all configurations with 95% confidence intervals (test split, seed 3000).

TABLE V
REPLICATION: MEAN F1 ON TWO INDEPENDENTLY GENERATED TEST SPLITS. THE OTSU FLOOR WAS 0.05 FOR SEED 2000 AND 0.12 (VALIDATION-SELECTED) FOR SEED 3000; M1–M3 DO NOT USE OTSU.

| Test split | M1 | M2 | M3 | M4 | M4 on `unchanged` |
|---|---|---|---|---|---|
| seed 2000 | 0.783 | 0.620 | 0.692 | 0.927 | 0.967 |
| seed 3000 | 0.797 | 0.620 | 0.702 | 0.933 | 1.000 |

to 0.12, and generated a fresh test split (seed 3000) to avoid reusing the old one. M4 then scores 1.000 on `unchanged`. Because the first split influenced this design decision, the seed-3000 results are the clean held-out evaluation; the differences in overall F1 between the two splits (0.927 vs. 0.933) are within the pair-to-pair variability and should not be read as an improvement.

### *E. Computational cost*

Registration dominates the extra cost. Across our two runs, pixel differencing needed 5–6 ms per pair, M3 33–46 ms and ATLAS 61–119 ms at $512 \times 384$, i.e. roughly 12–20 times the cost of M1 (absolute timings varied by about a factor of two between runs on a shared laptop, so only relative costs should be read from them). Fig. 4 shows the accuracy/runtime trade-off. On real photographs of up to 1024 px on the longest side, the full pipeline took 0.3–1.9 s per pair in the web interface. For offline or interactive use this is acceptable; for real-time video it would require optimisation.

### *F. Qualitative example*

Fig. 5 shows one `combined` pair. The baselines flag large parts of the image, mostly along edges displaced by the camera motion and across areas affected by the illumination gradient, whereas ATLAS isolates the inserted objects.

## VI. CASE STUDY ON REAL PHOTOGRAPHS

To see how the pipeline behaves outside the synthetic benchmark, we photographed three scenes with a handheld phone and processed them through the web interface (images downscaled to at most 1024 px). The parameters were those of the first tuning run, which coincide with Table II except for the Otsu floor of M4, which was 0.05 at that time instead of

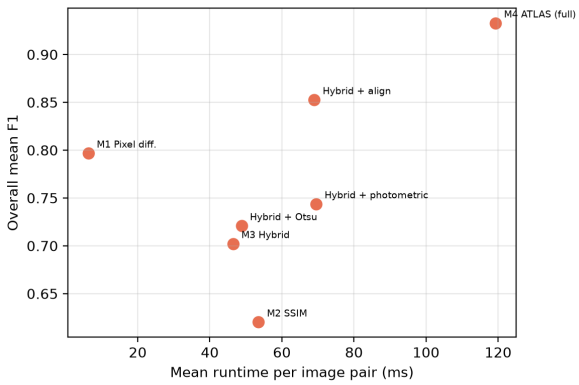



Fig. 4. Overall F1 versus mean runtime per image pair (test split, seed 3000).

TABLE VI
PERCENTAGE OF PIXELS MARKED AS CHANGED ON THREE REAL, UNANNOTATED PAIRS (M4 WITH THE INITIAL OTSU FLOOR OF 0.05). THE FLOOR MATTERS ONLY FOR THE ROOM PAIR, WHERE OTSU'S THRESHOLD WAS 0.063; ON THE PENGUIN AND CURTAIN PAIRS IT WAS 0.310 AND 0.467, ABOVE BOTH FLOORS.

| Pair | M1 | M2 | M3 | M4 |
|---|---|---|---|---|
| (a) Penguin | 37.13 | 72.84 | 54.73 | 62.68 |
| (b) Room | 0.67 | 44.09 | 11.50 | 5.12 |
| (c) Curtain | 34.84 | 78.47 | 52.12 | 29.50 |

the later validation-selected 0.12. The pairs are *not annotated*, so we report only the fraction of pixels each method marks as changed (Table VI) and inspect the overlays (Fig. 7). These three pairs are anecdotes, not a benchmark, and cannot support accuracy claims.

- **(a) Penguin:** a soft toy standing in front of a wall is removed, and the camera is moved back and rotated, so there is a large real change plus a large viewpoint change.
- **(b) Room:** two handheld shots of a room taken moments apart; no change was staged, and the pair differs mainly by a small viewpoint shift.
- **(c) Curtain:** a curtain is pulled aside, with a strong exposure change and a viewpoint shift.

Three observations follow. First, the SSIM-based methods mark very large areas on all pairs (M2: 44–78%), consistent with the low precision of SSIM on the benchmark (0.584). Second, the near-static room pair isolates the false-alarm behaviour: pixel differencing marks 0.67%, the fixed hybrid 11.50% and SSIM 44.09%, while M4 marks 5.12%, more than seven times as much as pixel differencing. Here Otsu chose a threshold of 0.063, close to the noise level, which is the same noise-splitting mechanism that produced false alarms on unchanged synthetic pairs (Section V-D). The validation-selected floor of 0.12 rules out thresholds that low, but we did not re-run this pair with it, so the 5.12% reflects the earlier floor. Third, on the penguin pair M4 marks 62.68% against 37.13% for pixel differencing, and visual inspection of the overlay shows its mask covering large wall areas that appear

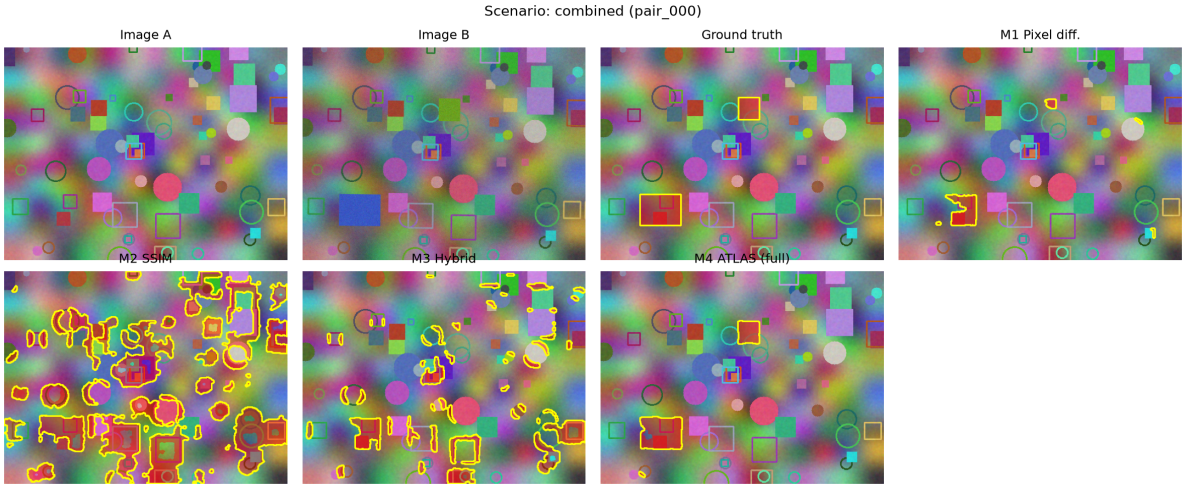



Fig. 5. A `combined` test pair: images $I_1$ and $I_2$, the ground truth, and the detections of M1–M4 (overlaid in red with a yellow contour).

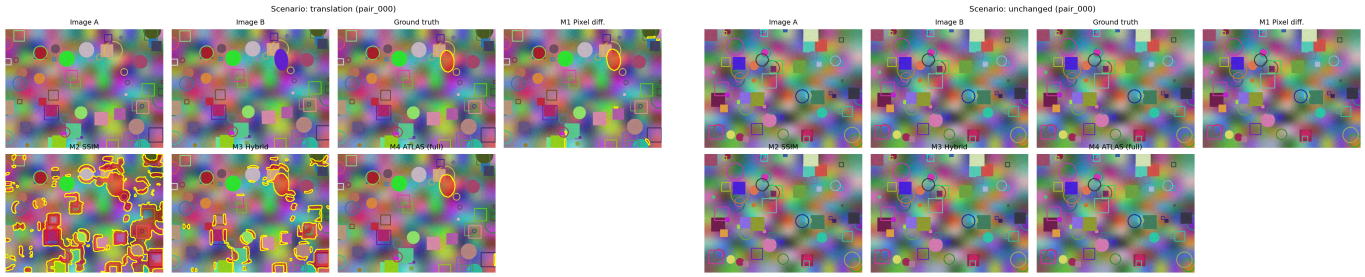



Fig. 6. Left: a translation pair. Right: an `unchanged` pair, on which no method should report change.

unchanged. We therefore cannot claim that ATLAS's synthetic advantage transfers to such scenes. On the curtain pair the true change is large, so the 29.5–78.5% range of the methods cannot be judged as error without annotations.

We hypothesise three causes, which we have not isolated experimentally: (i) *parallax*: a single homography can align at most one depth plane, but these scenes contain near and far surfaces; (ii) a *changed region covering a large part of the frame*, which violates the "mostly unchanged" assumption behind the global photometric model; and (iii) very large *exposure differences* that a global per-channel mapping cannot remove. Measuring these effects requires annotated real pairs, which we leave to future work.

## VII. Discussion and Limitations

**RQ1: fusion.** A fixed-weight fusion of pixel and SSIM maps did not improve on pixel differencing (F1 0.702 vs. 0.797). The structural term inherits SSIM's sensitivity to small offsets and noise, and the validation split pushed the weight to $\alpha = 0.75$ for every hybrid configuration. This is the upper edge of our grid, so the optimum may lie even closer to pixel-only; we did not search $\alpha \in (0.75, 1]$. The practical conclusion is that the benefit of ATLAS comes from the *preprocessing and the adaptive decision*, not from the fusion itself.

**RQ2: components.** Registration contributes the most ($+0.151$ over the fixed hybrid), photometric normalisation and the adaptive threshold add smaller gains, and the three together exceed the sum of their parts. The adaptive threshold is only safe with a floor.

**RQ3: failure cases.** On real photographs with parallax, large changed regions and strong exposure differences, the global geometric and photometric models do not hold, and ATLAS did not show a verifiable advantage over pixel differencing.

**Threats to validity.** (1) The benchmark was designed by us around the nuisance factors that ATLAS addresses (global rigid motion, global gain/bias/gradient), so the reported gains are an upper bound for real data. Its textured backgrounds also make feature matching easy. (2) Parameters were tuned on validation data pooled over all scenarios; per-scenario tuning would raise some baselines (e.g. pixel differencing under brightness change). (3) The Otsu floor was introduced after seeing the seed-2000 results; we mitigated this with a fresh test split. (4) F1 is averaged per image, which weights small and large changes equally; a pixel-pooled F1 would weight large changes more. The `unchanged` convention lifts all overall scores. (5) With 210 paired synthetic pairs the $p$-values are very small; effect sizes and the real-photo results are more informative about practical value. (6) We compare against classical baselines only, not against learned detectors, and timings come from a

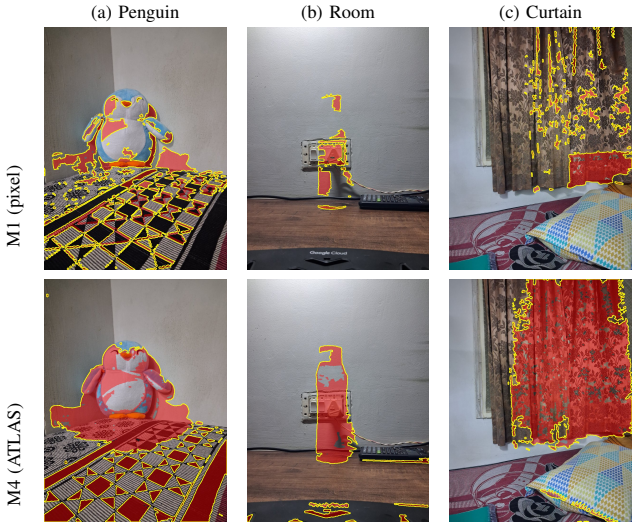



Fig. 7. Detections of M1 (top) and M4 (bottom) on the three real pairs, overlaid on the earlier image.

non-dedicated laptop.

**Future work.** The main step is a small annotated real dataset (or a LEVIR-CD subset [4]) to test the findings outside synthetic data. Methodologically, local or non-rigid registration (e.g. dense optical flow) should address parallax, a registration quality check could reject poor homographies instead of only counting inliers, illumination could be estimated from pixels judged unchanged rather than from the whole image, the fusion weight could be learned, and a comparison with Siamese networks [3] would place the classical pipeline relative to learned methods.

## VIII. Conclusion

We presented ATLAS, a training-free change detection pipeline combining registration, robust photometric normalisation, pixel/structural score fusion and floor-bounded Otsu thresholding, and evaluated it with a controlled, reproducible protocol. On a seven-scenario synthetic benchmark it reached a mean F1 of 0.933 on 210 held-out pairs, significantly above pixel differencing (0.797), a fixed hybrid (0.702) and SSIM (0.620), and the ranking replicated on a second test set. The ablation shows that registration matters most, that the components complement each other, and that fusing pixel and SSIM maps alone does not help. Real photographs with parallax and large exposure differences reveal the limits of global models, and we regard measuring and addressing them on annotated real data as the next step.

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